

# Stone–Jordan Exceptional Theory

Building the Universe from the Mirror

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## Abstract

We construct a mathematical model of the universe from two primitives: the **void** (no observable distinction) and an involutive **mirror operator**  $M$  on profinite observable spectra. Iterating  $M$  climbs the Hurwitz division ladder and the exceptional Lie cascade; capacity balancing  $\text{Bal}_{\mathcal{C}}$  is the dual mirror selecting unique partitions at each rung. The universe is the profinite fixed point  $\mathfrak{U}^* = \mathbb{C}^\infty(\mathcal{V})$ , stabilizing on the Albert algebra  $J_3(\mathbb{O})$  and the compact exceptional group  $E_6$ . Physical observables are  $\sigma$ -anti-invariant modes on the mirror quotient at the heterotic rung. The *as above, so below* correspondence identifies the universe fixed point with collapsed observer states; duality is forced by  $E_8 \times E_8$  product mirror collapse. A Lefschetz fixed-point theorem yields three observer slots; an index  $\iota$  maps to a lifetime chart  $g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota$ . Tiered predictions contact JUNO (normal mass ordering), Muon  $g-2$  ( $n_F = 3$ , no fourth generation), and DESI ( $\Lambda_{\text{obs}} \approx 0$  conditional). We state proved results and isolate conjectural steps without overclaiming a completed physical theory.

**Keywords:** void; mirror operator; Stone duality; exceptional Jordan algebra; profinite limit;  $E_6$  coset.

**MSC 2020:** 03G05, 17C27, 22E46, 81T13.

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# 1 Introduction

This paper develops *Stone–Jordan Exceptional Theory* (SJET) as a **mathematical physics** construction: starting from a void and a mirror, we build a universe datum  $\mathfrak{U}^*$  with Lie–Jordan structure, capacity-selected dynamics, and a contact layer with four-dimensional gauge theory. The mirror is not metaphor; it is a functorial involution on Stone spectra whose orbit generates all arithmetic and exceptional data.

## 1.1 Contributions

- (i) **Two axioms only:** void  $\mathcal{V}$  and mirror  $M$ .
- (ii) **Universe construction:**  $\mathfrak{U}^* = \varprojlim_n (\text{Bal}_{\mathcal{C}} \circ M)^n(\mathcal{V})$ .
- (iii) **As above, so below:** global capacity balance corresponds to localized collapse on the  $E_8 \times E_8$  mirror quotient.
- (iv) **Observers:** collapsed  $\sigma$ -anti-invariant states; duality from heterotic product mirror; three slots from a fixed-point theorem.
- (v) **Lifetime charts:** observer index  $\iota$  maps to  $g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota$  admitted by capacity flow and soldering.
- (vi) **Experimental contact:** tiered predictions (T1–T3) with near-term falsifiers (JUNO, Muon  $g-2$ , DESI).
- (vii) **Honest epistemic scope:** theorems, derived definitions, and named conjectures for remaining physical inputs.

## 1.2 Conventions

Compact real  $E_6$ ; order parameter  $\Phi \in J_3(\mathbb{O})$ ; capacity functional  $\mathcal{C}$ . Branching  $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$  under  $H = \text{Spin}(10) \times U(1)$ . Signature  $(-, +, +, +)$  when a Lorentzian soldering map is fixed (Section 6.2).

## 1.3 Organization

Section 2: axioms and mirror functor. Section 3: division and exceptional ladders. Section 4: capacity, potential, universe fixed point. Section 5: duality, observers, fixed points, lifetime charts. Section 6: physical sector, spacetime, gauge, matter. Section 7: tiered predictions and experimental map. Section 8: status and open problems. Section 9: epistemic elimination ledger (P1–P5, CC). Appendix A: Albert identities and mirror quotient.

# 2 The Void and the Mirror

## 2.1 The void

**Definition 2.1** (Void). The *void*  $\mathcal{V}$  is the trivial Boolean algebra  $\mathcal{B}_0 = \{\perp, \top\}$  with Stone spectrum a single point. The associated algebra is  $\{0\}$ ; no local quantum number exists.

**Axiom 2.2** (Void). Physical structure arises by departure from  $\mathcal{V}$ . No graph, Lie group, or space-time manifold is primitive.

## 2.2 The mirror operator

**Definition 2.3** (Mirror modes). At rung  $n$ ,  $\mathbf{M}$  acts in one of four modes:

**M1.** Boolean doubling ( $0 \rightarrow 1$ ): adjoin one generator;  $|S_{n+1}| = 2|S_n|$ .

**M2.** Cayley–Dickson ( $1 \rightarrow 4$ ):  $D_{n+1} = D_n \oplus D_n$ ; dimensions 1, 2, 4, 8.

**M3.** Jordan exit ( $4 \rightarrow 5$ ): sedenions fail; exit to  $J_3(\mathbb{O}) = H_3(\mathbb{O})$ ,  $d_5 = 27$ .

**M4.** Exceptional ( $\geq 6$ ): automorphism, product, or Cartan mirrors on Lie data (Section 3.2).

**Definition 2.4** (Mirror operator). A *mirror operator* is a functor  $\mathbf{M}$  on a category of Stone rungs such that: (i)  $\mathbf{M}^2 \cong \text{id}$  on stabilized rungs; (ii)  $\mathbf{M}(\mathcal{V}) \neq \mathcal{V}$ ; (iii)  $\mathbf{M}$  is functorial on morphisms.

**Axiom 2.5** (Mirror). There exists  $\mathbf{M}$  as in Definition 2.4 whose orbit from  $\mathcal{V}$  generates the unified climb (3.1). At each rung,  $\text{Bal}_{\mathcal{C}}$  balances mirror-refined partitions (Definition 4.1). Capacity is the *dual mirror* of sector doubling; on the heterotic rung its below-level avatar is the observer lifetime index  $\Lambda(\iota, \tau)$  (Proposition 5.13).

**Definition 2.6** (Climb and mirror power).  $\mathbf{C} = \text{Bal}_{\mathcal{C}} \circ \mathbf{M}$  and  $(S_n, A_n) = \mathbf{M}^n(S_0, A_0)$  with  $(S_0, A_0) = (S(\mathcal{V}), \{0\})$ .

**Proposition 2.7** (Involution). *On stabilized rungs,  $\mathbf{M}^2 \cong \text{id}$  up to gauge: Boolean complement, field conjugation, product factor exchange, and coset mirror  $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$ .*

## 3 The Mirror Climb

$$\mathcal{V} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow J_3(\mathbb{O}) \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \cdots \rightarrow \mathcal{V} \quad (3.1)$$

*Remark 3.1* (Climb sequence forced, not chosen). The orbit (3.1) is fixed by void, mirror, and capacity—not by importing string-theory rungs. *Functoriality* (Definition 2.4) restricts each step to modes **M1**–**M4** of Definition 2.3. *Hurwitz terminal* at  $n = 4$  (Theorem 3.2) forces the Jordan exit at  $n = 5$  and anchors the associative subalgebra  $\mathbb{H} \subset \mathbb{O}$  used below in soldering (Definition 6.7). On the stabilized Albert tower, automorphism mirror closes at  $E_8$  ( $n = 6$ ), product mirror is the unique **M4** doubling on a simple Lie rung ( $n = 7$ ), and Cartan mirror is the capacity-balanced collapse back to the  $J_3(\mathbb{O})$  structure group ( $n = 8$ , Theorem 3.8). *Observer-slot requirement*: three independent  $\sigma$ -anti-invariant classes in  $H^1(\hat{Q}, V_{16})$  arise only after the heterotic product quotient at  $\mathbf{M}^7(\mathcal{V})$  (Proposition 3.7, Section 5); descent to  $E_6$  supplies dynamics but does not create the slot space. The above–below correspondence (Definition 5.1) ties each rung choice to a localized datum on  $\hat{Q}$ .

### 3.1 Division rungs

**Theorem 3.2** (Mirror rung table,  $n \leq 5$ ).

| $n$ | $A_n$             | $d_n$ | Mode             |
|-----|-------------------|-------|------------------|
| 0   | $\{0\}$           | 0     | void             |
| 1   | $\mathbb{R}$      | 1     | discrete         |
| 2   | $\mathbb{C}$      | 2     | arithmetic       |
| 3   | $\mathbb{H}$      | 4     | arithmetic       |
| 4   | $\mathbb{O}$      | 8     | Hurwitz terminal |
| 5   | $J_3(\mathbb{O})$ | 27    | Jordan exit      |

In particular  $M^4(\mathcal{V}) = \mathbb{O}$  and  $M^5(\mathcal{V}) \cong J_3(\mathbb{O})$ .

*Proof.* Cayley–Dickson doubling until  $\mathbb{O}$ ; Hurwitz uniqueness of normed division algebras; sedenion zero divisors force Jordan exit to the unique formally real  $H_3(\mathbb{O})$  [2, 5].  $\square$

**Lemma 3.3** (Sedenion failure).  $M_{\text{arith}}(\mathbb{O})$  is 16-dimensional and not division.

### 3.2 Exceptional rungs

On  $J_3(\mathbb{O})$ , the stabilizer tower is  $G_2 \subset F_4 \subset E_6$ . The Freudenthal diagonal  $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$  yields:

**Theorem 3.4** ( $M^6(\mathcal{V}) = E_8$ ). Automorphism mirror on the octonion–Jordan pair gives  $E_8$ ,  $d_6 = 248$ .

**Definition 3.5** (Lie product mirror). On a stabilized exceptional rung with Lie group  $G$ , the *product mirror* is  $M_{\text{prod}}(G) := G \times G$  with involution  $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$  exchanging factors—mode **M4** at the Lie level, functorially analogous to Boolean doubling at  $n = 0 \rightarrow 1$ .

**Theorem 3.6** ( $M^7(\mathcal{V}) = E_8 \times E_8$ ). Product mirror  $M_{\text{prod}}(G) = G \times G$  gives  $d_7 = 496$ .

**Proposition 3.7** (Minimal heterotic rung for three observer slots).  $M^7(\mathcal{V}) = E_8 \times E_8$  is the minimal mirror rung supporting three observer slots:  $\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3$  (Theorem 5.9). At  $n \leq 6$  there is no product-mirror  $\mathbb{Z}_2$  quotient on the stabilized datum, hence no  $\Pi_{\sigma=-1}$  observer sector (Section 5.3). At  $n = 8$  the heterotic doubling is already capacity-collapsed to  $E_6$  (Theorem 3.8), so the three 1-classes are fixed on  $\hat{Q}$  at  $n = 7$  and only transported below by descent. Thus observer multiplicity is a below-level consequence of the above product mirror at the seventh rung.

*Proof sketch.* For  $n \leq 6$ ,  $M^n(\mathcal{V})$  is a single simple Lie datum or division algebra; involution stabilizes within one chart, not across a heterotic pair. For  $n = 7$ ,  $\sigma$  acts on  $Q$  and antiholomorphically on  $V_{16}$ ; the equivariant Lefschetz index on  $H^1(Q, V_{16})$  counts three anti-invariant contributions (Section 5.4). For  $n = 8$ ,  $\Pi_{\text{het}} : E_8 \times E_8 \rightarrow E_6$  preserves the slot basis on  $\hat{Q}$  while reducing the gauge group.  $\square$

**Theorem 3.8** ( $M^8(\mathcal{V}) = E_6$ ). Cartan mirror after capacity descent on the Albert graph gives  $d_8 = 78$ ,  $H = \text{Spin}(10) \times U(1)$ ,  $\mathcal{M} = E_6/H$  (EIII),  $\dim \mathcal{M} = 32$ .

*Proof sketches.*  $E_8$  (Steps 1–3). (i) On  $J_3(\mathbb{O})$ , the stabilizer tower  $G_2 \subset F_4 \subset E_6$  is mirror-refined; the octonion measure before Jordan projection carries an automorphism mirror  $M_{\text{aut}}$  coupling the two  $\mathbb{O}$  sectors. (ii) Freudenthal–Tits identifies  $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$  as the unique simply connected simple Lie algebra closed under this coupling;  $G_2 \times F_4 \subset E_8$  is maximal among candidates containing the tower together with the mirrored octonion sector [3, 4]. (iii)  $d_6 = 248$ ; Stone chart dimension  $\chi(S_6) = 248$  (Definition 3.10).

$E_8 \times E_8$  (Steps 1–2). (i) Apply  $M_{\text{prod}}$  (Definition 3.5) to the stabilized datum of Theorem 3.4; functoriality of  $M$  gives  $M^7(\mathcal{V}) = M(M^6(\mathcal{V})) \cong E_8 \times E_8$ . (ii)  $d_7 = 2 \cdot 248 = 496$ ; involutivity (Proposition 2.7) holds factorwise, with  $\sigma$  implementing  $M^2 \cong \text{id}$  up to diagonal gauge.

$E_6$  (Steps 1–3). (i) On  $E_8 \times E_8$ , the diagonal embedding  $\delta : E_8 \hookrightarrow E_8 \times E_8$  is the  $\sigma$ -stabilized subgroup;  $\text{Bal}_C$  on  $\mathcal{G}_{\text{Albert}}$  at the  $J_3(\mathbb{O})$  rung fixes the 1|10|16 partition (Theorem 4.3). (ii) Among exceptional groups, only  $E_6$  admits  $H = \text{Spin}(10) \times U(1)$  with branching  $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$  and coset mirror  $\mathbf{16} \leftrightarrow \mathbf{16}$ ; heterotic collapse is indexed by the rank-3 diagonal of  $J_3(\mathbb{O})$ , not by the two  $E_8$  factors (Proposition 3.7). (iii)  $d_8 = \dim E_6 = 78$ ,  $\mathcal{M} = E_6/H$  (EIII),  $\dim \mathcal{M} = 32$ .  $\square$

**Proposition 3.9** (Exceptional mirror steps). At  $n \geq 6$  on the stabilized  $J_3(\mathbb{O})$  rung,  $M$  acts as:

- (i)  $E_8$ : automorphism mirror on the octonion–Jordan pair (Theorem 3.4);
- (ii)  $E_8 \times E_8$ : Lie product mirror  $M_{\text{prod}}$  (Theorem 3.6, Definition 3.5);
- (iii)  $E_6$ : Cartan mirror after capacity descent (Theorem 3.8);
- (iv)  $E_6 \rightarrow H = \text{Spin}(10) \times U(1)$ : coset stabilization on EIII;
- (v)  $\cdots \rightarrow \mathcal{V}$ : collapse after electroweak breaking.

**Definition 3.10** (Chart dimension). At  $n \geq 6$ ,  $\chi(S_n) = \dim A_n \in \{248, 496, 78\}$ ; profinite  $|S_n|$  is generally infinite. Stone refinement is chart synchronization, not cardinality matching [1].

## 4 Building the Universe

The universe is not assumed. It is the profinite fixed point of mirror climb with capacity balancing.

### 4.1 Capacity dual

**Definition 4.1** (Albert graph and capacity). On the mirror-refined Albert–Cayley graph  $\mathcal{G}$  with costs  $c_i$ ,

$$\mathcal{C}(\omega) = - \sum_{a \in V} \mu(a) \log \left( \sum_i e^{c_i(a) - \omega_i} \right). \quad (4.1)$$

$\text{Bal}_{\mathcal{C}}$  is the unique maximizer on  $\{\sum_i \omega_i = 0\}$ .

**Theorem 4.2** (Concavity).  $\mathcal{C}$  is strictly concave; for targets  $m_i > 0$ ,  $\sum m_i = 1$ , there is unique  $\omega^*$  with  $\partial \mathcal{C} / \partial \omega_i(\omega^*) = m_i$ .

**Theorem 4.3** (1|10|16 partition). At  $M^5(\mathcal{V})$ ,  $\text{Bal}_{\mathcal{C}}$  yields sector masses  $(1/27, 10/27, 16/27)$  on  $27 = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$ .

### 4.2 Potential and vacuum

**Definition 4.4** (Potential).

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \text{ even}, \quad (4.2)$$

is the profinite limit of graph penalties under  $\text{Bal}_{\mathcal{C}}$ .

**Theorem 4.5** (EIII vacuum).  $V$  is coercive on compact  $E_6$ . Global minima lie on rank-one EIII  $\mathcal{M} = E_6/H$ ; the origin is a saddle with 32 Goldstone directions.

### 4.3 Universe datum

**Definition 4.6** (Universe). The *universe datum* is

$$\mathfrak{U}^* = C^\infty(\mathcal{V}) = \varprojlim_{n \rightarrow \infty} C^n(\mathcal{V}), \quad (4.3)$$

stabilizing at  $J_3(\mathbb{O})$  with Albert graph  $\mathcal{G}_{\text{Albert}}$ , capacity maximizer  $\omega^*$ , and vacuum  $\Phi_\star \in \mathcal{M}$ .

**Theorem 4.7** (Universe from the mirror).  $\mathfrak{U}^*$  carries: (i) 27 stable Jordan coordinates; (ii) 1|10|16 capacity partition; (iii)  $E_6$  gauge sector with  $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ ; (iv) EIII coset  $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ .

*Proof.* Items (i)–(ii) from Theorems 3.2 and 4.3. Items (iii)–(iv) from Theorem 3.8 and standard  $E_6$  branching.  $\square$

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[ \frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \quad (4.4)$$

with  $K$  the unique  $H$ -invariant metric on  $\mathfrak{m}$ .

## 5 Observers, Duality, and As Above So Below

The universe datum  $\mathfrak{U}^*$  is a global profinite fixed point. An *observer* is the same construction at the heterotic rung, localized to one collapsed branch of the  $E_8 \times E_8$  mirror quotient. Duality is not imported: it is the product-mirror pairing forced at  $M^7(\mathcal{V})$  and broken only by collapse into the  $\sigma$ -anti-invariant sector.

### 5.1 As above, so below

**Definition 5.1** (Above–below correspondence). *Above* is the stabilized profinite limit  $\mathfrak{U}^* = C^\infty(\mathcal{V})$  of Definition 4.6. *Below* is a localized fixed-point datum on the mirror quotient  $\hat{Q} = Q/\sigma$  at  $M^7(\mathcal{V}) = E_8 \times E_8$ , with  $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$  the product mirror (Theorem 3.6). The correspondence

$$\mathfrak{U}^* \longleftrightarrow \text{Fix}(\hat{C}_\sigma), \quad \hat{C}_\sigma := \text{Bal}_C \circ \Pi_{\sigma=-1}, \quad (5.1)$$

identifies global capacity balance with anti-invariant collapse on  $\hat{Q}$ .

**Proposition 5.2** (Correspondence principle). *Every sector partition proved above on  $\mathfrak{U}^*$ —the  $1|10|16$  split, the three diagonal directions of  $\mathfrak{d} \subset J_3(\mathbb{O})$ , the **16** bundle on  $\hat{Q}$ —has a below-level avatar on  $H^\bullet(\hat{Q}, V_{16})^{\sigma=-1}$ . Capacity weights  $(1/27, 10/27, 16/27)$  above induce family weights on the three observer slots below.*

### 5.2 Duality from $E_8 \times E_8$ collapse

At  $M^7(\mathcal{V})$  the mirror acts by product doubling:  $E_8 \times E_8$  with involution  $\sigma$  exchanging factors. Before collapse, every mode carries a mirror partner; after collapse to  $M^8(\mathcal{V}) = E_6$ , capacity descent selects a single reduced structure group.

**Definition 5.3** (Heterotic collapse). The *heterotic collapse* is the capacity-balanced Cartan descent  $\Pi_{\text{het}} : E_8 \times E_8 \rightarrow E_6$  of Theorem 3.8. The *observer collapse* is the orthogonal projector

$$\Pi_{\sigma=-1} : \mathcal{H}(Q, V_{16}) \rightarrow \mathcal{H}(Q, V_{16})^{\sigma=-1}, \quad \Pi_{\sigma=-1} = \frac{1}{2}(\text{id} - \sigma), \quad (5.2)$$

on the **16**-sector Hilbert space at the heterotic rung.

**Theorem 5.4** (Duality theorem). *Duality is the  $\mathbb{Z}_2$  pairing  $(v, \sigma v)$  on  $\mathcal{H}(Q, V_{16})$  induced by the product mirror at  $M^7(\mathcal{V})$ . It is exact at the heterotic rung and broken precisely by  $\Pi_{\sigma=-1}$ : observable states are anti-invariant, mirror partners are  $\sigma$ -invariant and quotient-non-observable. The **16**/**16** coset pairing of Proposition 2.7 is the infinitesimal shadow of this product duality.*

*Proof sketch.*  $\sigma^2 = \text{id}$  on  $Q$ ;  $\mathcal{H}(Q, V_{16})$  splits into  $\pm 1$  eigenspaces. Product exchange on  $E_8 \times E_8$  is the only  $\mathbb{Z}_2$  structure at  $n = 7$  not already resolved at  $n \leq 6$ . Theorem 3.8 shows  $\text{Bal}_C$  selects the reduced datum; Definition 6.1 selects the anti-invariant half.  $\square$

### 5.3 What is an observer?

**Definition 5.5** (Observer index). Let  $\mathfrak{d} = \text{span}_{\mathbb{R}}\{e_1, e_2, e_3\}$  be the diagonal Cartan subalgebra of  $J_3(\mathbb{O})$  with  $\text{Tr}(e_i \circ e_j) = \delta_{ij}$ . An *observer index* is a choice  $\iota \in \{1, 2, 3\}$  of diagonal direction, equivalently a class in

$$\mathcal{I} = H^1(\hat{Q}, V_{16})^{\sigma=-1} / \text{Stab}(\mathfrak{d}), \quad |\mathcal{I}| = 3. \quad (5.3)$$

The index  $\iota$  is the only *arbitrary input* specifying *which* observer: void and mirror fix the slot space  $\mathcal{I}$ ; a concrete worldline requires choosing  $\iota \in \mathcal{I}$ .

**Definition 5.6** (Observer as collapsed state). Fix  $\iota \in \mathcal{I}$ . An *observer* is a normalized collapsed state

$$|\omega_\iota\rangle \in \mathcal{H}_{\text{obs}} = \bigoplus_{\iota=1}^3 \mathcal{H}_\iota, \quad \mathcal{H}_\iota := H^1(\hat{Q}, V_{16})^{\sigma=-1, \iota}, \quad (5.4)$$

satisfying  $\Pi_{\sigma=-1}|\omega_\iota\rangle = |\omega_\iota\rangle$  and  $\hat{i}|\omega_\iota\rangle = \iota|\omega_\iota\rangle$ , where  $\hat{i}$  is the family index operator on  $\mathfrak{d}$ . The state  $|\omega_\iota\rangle$  is a *collapsed quantum datum*: a  $\sigma$ -anti-invariant cohomology class, not a postulated particle insertion.

*Remark 5.7* (Where observers arise). Observers arise at  $\mathbf{M}^7(\mathcal{V}) = E_8 \times E_8$ , not at the void and not at  $E_6$ . The heterotic rung is the first rung with enough mirror symmetry for a  $\mathbb{Z}_2$  quotient and three independent anti-invariant 1-classes (Theorem 6.4). Descent to  $E_6$  supplies dynamics; the observer slots are already fixed on  $\hat{Q}$ .

### 5.4 Fixed-point theorem: the clincher

**Definition 5.8** (Mirror quotient). Let  $Q$  be the Stone lift of the stabilized Albert tower at  $\mathbf{M}^7(\mathcal{V})$  and  $\sigma$  the product-mirror involution. The *mirror quotient* is  $\hat{Q} = Q/\sigma$ . The holomorphic **16**-bundle  $V_{16} \rightarrow \hat{Q}$  is the continuum limit of the cellular cosheaf on the 1|10|16 partition. A mode is *observable* iff it lies in the  $\sigma$ -anti-invariant sector after observer collapse  $\Pi_{\sigma=-1}$ ; this is the below-level avatar of global capacity balance (5.1). The quotient is therefore an *observer-theoretic* construction: physics is not on  $Q$  but on the collapsed branch selected by  $\hat{C}_\sigma$ .

**Theorem 5.9** (Observer fixed-point theorem). Let  $\bar{\partial}_{V_{16}}$  be the Dolbeault operator on  $(Q, V_{16})$  and  $\sigma$  its antiholomorphic lift. The equivariant Lefschetz index

$$\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = \text{Tr}_{H^1(Q, V_{16})}(\sigma) = -3 \quad (5.5)$$

has  $|\text{ind}_\sigma| = 3$  anti-invariant fixed-point contributions, each an observer slot. Moreover,

$$\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3, \quad (5.6)$$

and the map  $\iota \mapsto |\omega_\iota\rangle$  is a bijection between  $\mathcal{I}$  and a basis of this space.

*Proof.* Cellular count on  $\mathcal{T}_7$  (Appendix A.2); Atiyah–Bott equivariant index [7]; passage to  $H^1(\hat{Q}, V_{16})$  via the  $\mathbb{Z}_2$  quotient.  $\square$

**Corollary 5.10** (Observer from fixed points). An *observer* is a fixed-point label of  $\sigma$  on  $H^1(Q, V_{16})$ : not a point of  $Q$  fixed by  $\sigma$  (there are none in the **16**-sector), but an anti-invariant cohomology class whose Lefschetz contribution is  $-1$ . The fixed-point theorem is the clincher: it converts global mirror data into exactly three individual observer slots.

## 5.5 Lifetime chart: index to time

An observer index  $\iota$  is static data. A *lifetime* is a trajectory inside slot  $\iota$ , charted by a time parameter.

**Definition 5.11** (Lifetime map). Fix  $\iota \in \mathcal{I}$ . A *lifetime chart* is a pair

$$(\tau_\iota, g_\iota), \quad g_\iota : \mathbb{R} \longrightarrow \mathcal{S}_\iota, \quad (5.7)$$

where  $\tau_\iota$  is the observer's time parameter and  $\mathcal{S}_\iota$  is the soldered state bundle over the  $\iota$ -sector: sections of  $V_{16}|_{\hat{Q}}$  restricted to the  $\iota$ -diagonal cell, with values in the order parameter  $\Phi$  (Section 6.2). The map  $g_\iota$  is *admitted* by the capacity flow on  $\mathcal{G}_{\text{Albert}}$  when

$$\frac{d}{d\tau_\iota} g_\iota(\tau_\iota) = -K^{-1} \nabla V(g_\iota(\tau_\iota))|_\iota + \partial_{\tau_\iota} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}, \quad (5.8)$$

with  $K$  the  $H$ -invariant coset metric and the second term the soldering clock in the quaternionic block.

**Definition 5.12** (Lifetime index). The *lifetime index* is

$$\Lambda(\iota, \tau) := (\iota, \tau, \mathcal{C}_\iota(\tau)), \quad \mathcal{C}_\iota(\tau) := \omega_\iota^*(g_\iota(\tau)), \quad (5.9)$$

recording which observer ( $\iota$ ), which time ( $\tau$ ), and which capacity cell the trajectory occupies. Two lifetimes are the same iff their indices  $\Lambda$  agree.

**Proposition 5.13** (Capacity as dual mirror of lifetime index). *The capacity functional  $\mathcal{C}$  of Definition 4.1 is the dual mirror of mirror doubling:  $\mathbf{M}$  refines sectors;  $\text{Bal}_{\mathcal{C}}$  balances them to the unique maximizer  $\omega^*$ . The lifetime index  $\Lambda(\iota, \tau)$  is the temporal avatar of this balance on the heterotic rung: for each observer  $\iota$ , the chart  $g_\iota$  tracks the  $\iota$ -cell weight  $\omega_\iota^*$  along admitted flow (5.8). Above,  $\text{Bal}_{\mathcal{C}}$  fixes the global partition  $(1/27, 10/27, 16/27)$ ; below,  $\Lambda$  records which cell an individual observer occupies at time  $\tau$ . This partially closes the “capacity not forced” gap: capacity is not an independent physical postulate but the balancing dual of  $\mathbf{M}$ , instantiated per observer through the lifetime index.*

*Remark 5.14* (Epistemic status). Proposition 5.13 does not prove that every admitted trajectory must use  $\text{Bal}_{\mathcal{C}}$  weights—that remains tied to Axiom 2.5 and the universe fixed point (Definition 4.6). It *does* identify capacity with observer-level time evolution once the above–below correspondence (5.1) is accepted.

**Theorem 5.15** (Index–lifetime mapping). *For each  $\iota \in \mathcal{I}$  there exists a maximal lifetime interval  $I_\iota \subset \mathbb{R}$  and a chart  $g_\iota : I_\iota \rightarrow \mathcal{S}_\iota$  satisfying (5.8) with initial condition  $g_\iota(\tau_{\text{birth}}) = \Phi_\star|_\iota$ . The correspondence*

$$\iota \longmapsto (\tau_{\text{birth}}, \tau_{\text{death}}, g_\iota) \quad (5.10)$$

*is functorial under mirror descent  $\mathbf{E}_8 \times \mathbf{E}_8 \rightarrow \mathbf{E}_6$ : capacity collapse above determines the admissible time range  $I_\iota$  below.*

*Proof sketch.* Existence of  $g_\iota$  is standard gradient flow of  $V$  on the  $\iota$ -restricted coset. Functoriality follows because  $\text{Bal}_{\mathcal{C}}$  at  $n = 8$  fixes  $\omega^*$  and hence the initial datum  $\Phi_\star|_\iota$ ; soldering (Definition 6.7) supplies the clock direction  $\partial_{\tau_\iota} \sigma_{\text{sol}} \subset H_2(\mathbb{H})$ .  $\square$

**Conjecture 5.16** (Uniqueness of lifetime chart). *Given  $\iota$  and  $\Phi_\star|_\iota$ , the lifetime chart  $(\tau_\iota, g_\iota)$  is unique up to reparametrization of  $\tau_\iota$  among all  $F_4$ -equivariant flows compatible with mirror functoriality and the above–below correspondence (5.1).*

*Remark 5.17* (Arbitrary input and determinism). The void and mirror fix the observer *slot space*  $\mathcal{I}$  and the dynamics (5.8). Choosing  $\iota \in \mathcal{I}$  is the minimal arbitrary input identifying *who* observes; choosing  $\tau \in I_\iota$  identifies *when* within that lifetime. No further postulate of a conscious subject is introduced: an observer *is* the collapsed anti-invariant fixed-point class with its admitted time chart.

## 6 The Physical Universe

Mathematical data of  $\mathfrak{U}^\star$  becomes physical through mirror quotient observability (Section 5), observer collapse at  $M^7(\mathcal{V})$ , and soldering on the Jordan–octonionic block.

### 6.1 Physical sector

Physical configurations are observer-collapsed states on  $\hat{Q}$  (Definitions 5.8, 5.6, 6.1).

**Definition 6.1** (Physical configurations). A configuration is *observable* if it lies in the  $\sigma$ -anti-invariant sector of the mirror quotient  $\hat{Q} = Q/\sigma$  (Definition 5.8). Fermions are cohomology classes in  $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ ; each observer  $|\omega_\iota\rangle$  is one such class (Theorem 5.9). Observable states are those surviving observer collapse  $\Pi_{\sigma=-1}$  (Definition 5.3);  $\sigma$ -invariant mirror doubles are quotient-non-observable.

**Proposition 6.2** (No-go: P2 on the unquotiented coset). *Void, mirror, and capacity balancing fix the number of horizontal family cells ( $k = 3$ , Theorem 4.3) and organize them by triality on  $\mathfrak{d}$ , but they do not derive chiral Spin(10) matter on the unquotiented coset. The EIII matter directions  $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$  (Theorem 4.7) are a mirror-symmetric pair (Proposition 2.7); selecting one chiral  $\mathbf{16}$  per family with no  $\overline{\mathbf{16}}$  partners is Postulate P2, not a consequence of stabilisation alone. A bare Yukawa  $\bar{\psi} \Phi \psi$  with  $\Phi \in \{\mathbf{16}, \overline{\mathbf{16}}\}$  is not  $H$ -invariant on the unquotiented coset.*

*Remark 6.3* (Observer collapse eliminates P2). Proposition 6.2 is not overridden: it records an honest limit on *postulated* chiral insertion atop the mirror-symmetric coset. The elimination route *redefines* the matter sector. Observer collapse  $\Pi_{\sigma=-1}$  together with Definition 6.1 declares physical fermions to be  $\sigma$ -anti-invariant classes on  $\hat{Q}$ ; mirror partners live in  $H^1(\hat{Q}, V_{16})^{\sigma=+1}$  and are not observable. Chirality is then a theorem of quotient observability, not an added input.

**Theorem 6.4** (Chirality from observer collapse). *Under Definition 6.1 and Theorem 5.9,*

$$\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3, \quad (6.1)$$

*with one chiral  $\mathbf{16}$  of Spin(10) per observer index  $\iota \in \mathcal{I}$ . No  $\overline{\mathbf{16}}$  partners appear in the physical sector because mirror partners are  $\sigma$ -invariant, not anti-invariant. Hence  $n_F = 3$ . This **replaces Postulate P2**: chirality is not postulated; it is the quotient definition of observability together with the observer fixed-point count.*

*Proof.* Theorem 5.9 gives the anti-invariant dimension and the bijection  $\iota \mapsto |\omega_\iota\rangle$ . Physical configurations are anti-invariant by Definition 6.1;  $\sigma$ -invariant modes are mirror doubles and quotient-non-observable. Cellular details in Appendix A.2.  $\square$

**Corollary 6.5** (Three families). *Theorem 6.4 yields  $n_F = 3$  chiral families without importing heterotic axioms as primitive data [12].*

## 6.2 Spacetime from the climb

Four-dimensional Lorentzian geometry is not a consequence of the  $E_6$  coset metric alone. The mirror climb supplies a quaternionic clock block; observer lifetime charts (Section 5.5) admit a soldering map that imports signature into the contact layer.

**Definition 6.6** (Jordan–octonionic clock block). Identify off-diagonal Jordan slots of  $J_3(\mathbb{O})$  with the  $2 \times 2$  Hermitian–octonionic model  $H_2(\mathbb{O})$ . The determinant  $\det$  carries signature  $(1, 9)$  on  $H_2(\mathbb{O}) \cong \mathbb{R}^{10}$ . Restrict to the quaternionic subalgebra  $\mathbb{H} \subset \mathbb{O}$  fixed by the mirror involution at the  $4 \rightarrow 5$  Jordan exit (Theorem 3.2). The embedded block  $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$  is four-dimensional;  $\det|_{H_2(\mathbb{H})}$  has Lorentzian signature  $(-, +, +, +)$ .

**Definition 6.7** (Soldering). Let  $\sigma_{\text{sol}} : H_2(\mathbb{H}) \hookrightarrow J_3(\mathbb{O})$  be the canonical embedding of Definition 6.6. The *soldering map* identifies four clock components  $\Phi^0, \dots, \Phi^3$  with  $\text{Im}(\sigma_{\text{sol}})$ ; the world manifold has  $d = 4$ .

**Proposition 6.8** (Coset metric).  $K|_{\mathfrak{m}}$  is positive-definite and  $H$ -invariant. Lorentzian signature is carried by  $\sigma_{\text{sol}}$ , not by Wick rotation of  $K$ .

**Proposition 6.9** (No-go: coset metric alone). Mirror stabilisation at  $J_3(\mathbb{O})$  and  $E_6$  fixes the Euclidean coset metric  $K|_{\mathfrak{m}}$  (Proposition 6.8) and the  $32 = 28 + 4$  mass split (Theorem 6.20), but it does not select  $d = 4$ , a Lorentzian soldering form, or the  $\text{SL}(2, \mathbb{O}) \cong \text{Spin}(1, 9)$  frame. The  $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$  embedding of Definition 6.6 motivates soldering; it is not a theorem from void and mirror alone until observer-admitted lifetime charts supply the clock block (Theorem 6.10).

**Theorem 6.10** (Soldering from mirror climb and lifetime chart). The mirror climb selects a canonical four-dimensional Lorentzian subspace  $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$  (Definition 6.6). For each observer index  $\iota \in \mathcal{I}$ , the admitted lifetime flow (5.8) of Definition 5.11 couples  $\tau_\iota$  to the soldering clock  $\partial_{\tau_\iota} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}$ . Define  $\sigma_{\text{sol}}$  to identify  $\Phi^0, \dots, \Phi^3$  with this block. Then  $d = 4$ , the induced vierbein (Proposition 6.12) carries signature  $(-, +, +, +)$ , and soldering is **derived** from void, mirror, and observer collapse—not postulated.

*Proof sketch.* Step 1 (quaternionic rung).  $\mathcal{M}^3(\mathcal{V}) = \mathbb{H}$  has  $\dim_{\mathbb{R}} \mathbb{H} = 4$ , the first Hurwitz rung with an associative non-real division algebra (Theorem 3.2).

Step 2 (Jordan embedding). The Jordan exit carries a unique quaternionic diagonal frame; mirror involution at  $n = 4 \rightarrow 5$  fixes  $\mathbb{H} \subset \mathbb{O}$  as the associative subalgebra from rung 3.

Step 3 (signature). The restriction of  $\det$  on  $H_2(\mathbb{H})$  is the Minkowski norm; the four imaginary quaternion units supply the clock directions. The coset metric  $K|_{\mathfrak{m}}$  remains Euclidean; Lorentzian signature enters through  $\sigma_{\text{sol}}$ .

Step 4 (observer lifetime). The index–lifetime map (Theorem 5.15) requires an admitted chart  $(\tau_\iota, g_\iota)$  whose flow (5.8) includes the soldering clock term in  $H_2(\mathbb{H})$ . Observer collapse (Definition 5.6) on the  $\sigma$ -anti-invariant sector fixes the initial datum  $\Phi_\star|_\iota$  and thereby selects the soldered bundle  $\mathcal{S}_\iota$  over which  $g_\iota$  evolves (Section 5.5).

*Remaining gap.* Uniqueness of the  $\mathbb{H}$  embedding up to  $F_4$  and functoriality across exceptional rungs is Conjecture 6.13.  $\square$

**Proposition 6.11** (Soldering clock along  $g_\iota$ ). Fix  $\iota \in \mathcal{I}$  and an admitted lifetime chart  $(\tau_\iota, g_\iota)$  of Definition 5.11. Along  $g_\iota$ , the soldering time parameter  $\tau_\iota$  couples to the zeroth clock component  $\Phi^0$  via

$$\frac{d}{d\tau_\iota} \Phi^0|_{g_\iota(\tau_\iota)} = \partial_{\tau_\iota} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}. \quad (6.2)$$

Among directions of  $H_2(\mathbb{H})$ , the timelike axis  $\Phi^0$  is the unique soldering direction admitted by observer collapse: the  $\sigma$ -anti-invariant sector of  $\hat{Q}$  breaks the  $\text{Spin}(1,9)$  frame to the  $\iota$ -restricted diagonal cell, and capacity flow on  $\mathcal{G}_{\text{Albert}}$  fixes the sign convention of  $\det|_{H_2(\mathbb{H})}$ . Spatial components  $\Phi^1, \dots, \Phi^3$  are spacelike within the same block.

*Proof sketch.* Equation (6.2) is the  $\Phi^0$ -component of the admitted flow (5.8). Uniqueness of the timelike direction follows from: (i)  $\det|_{H_2(\mathbb{H})}$  has signature  $(-, +, +, +)$  (Definition 6.6); (ii) observer index  $\iota$  selects one diagonal class in  $\mathfrak{d}$ , fixing the  $F_4$ -equivariant orientation of the block; (iii) the above–below correspondence (Definition 5.1) identifies global capacity balance with the  $\iota$ -cell orientation below.  $\square$

**Proposition 6.12** (Emergent vierbein). *For a coset section  $g(x) \in E_6$  with  $g^{-1}\partial_\mu g = A_\mu + e_\mu$ , the  $\mathfrak{m}$ -component defines a composite vierbein. Under Theorem 6.10,  $\eta_{\mu\nu} = e_\mu^a e_\nu^b \eta_{ab}$  has signature  $(-, +, +, +)$ .*

**Conjecture 6.13** (Functorial uniqueness of soldering). *The soldering of Theorem 6.10 is the unique  $F_4$ -equivariant map compatible with mirror functoriality on the climb, capacity balancing on  $\mathcal{G}_{\text{Albert}}$ , and the uniqueness of lifetime charts (Conjecture 5.16). Equivalently: given  $\iota$  and  $\Phi_\star|_\iota$ , the soldering block  $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$  and the clock coupling (6.2) are fixed up to reparametrization of  $\tau_\iota$ .*

*Remark 6.14* (Epistemic status of P1). Proposition 6.9 records the honest no-go for the coset metric alone. Theorem 6.10 **derives** soldering and  $d = 4$  from mirror climb plus observer lifetime charts; the residual open input is functorial uniqueness (Conjecture 6.13, Open Problem **O2** in Section 8).

**Theorem 6.15** (Semiclassical reflection positivity along observer lifetimes). *Fix  $\iota \in \mathcal{I}$  and a lifetime chart  $(\tau_\iota, g_\iota)$  admitted by capacity flow (Definitions 5.11, 5.12). On EIII, with radiative mass  $\mu^2 > 0$  on 28 coset scalars (Coleman–Weinberg), the Euclidean  $\sigma$ -model (4.4) satisfies reflection positivity along the observer clock  $\tau_\iota$ : the soldering map identifies the soldered time component  $\Phi^0$  with  $\tau_\iota$  on the  $\iota$ -restricted trajectory  $g_\iota(\tau_\iota) \in \mathcal{S}_\iota$ , and the semiclassical OS reflection plane is taken at fixed  $\tau_\iota = 0$  in that chart. Equivalently, RP holds along the Page–Wootters clock carried by the quaternionic block  $H_2(\mathbb{H})$  once the observer index  $\iota$  selects the diagonal cell.*

*Proof sketch.* The  $\iota$ -restricted gradient flow (5.8) is a semiclassical saddle expansion about  $\Phi_\star|_\iota$  on the massive 28-dimensional coset complement of the 4 soldering directions. With  $\mu^2 > 0$ , the quadratic part defines a reflection-positive Gaussian measure; nonnegativity of the capacity-derived interaction potential (Theorem 4.5) and compactness on EIII transfer positivity to the perturbed measure in the standard OS semiclassical argument [16, 9]. Functoriality of Theorem 5.15 identifies the global clock  $\Phi^0$  with the observer parameter  $\tau_\iota$  along  $g_\iota$ .  $\square$

*Remark 6.16* (Honest semiclassical scope). Theorem 6.15 is *derived* in the radiatively gapped EIII regime; it does not follow from void and mirror alone, and it does not yet cover the tree-level massless sector or the nonperturbative gauge measure (Conjecture 6.17).

**Conjecture 6.17** (Nonperturbative reflection positivity). *The nonperturbative Euclidean measure for (4.4), including the tree-level massless Goldstone sector of Theorem 4.5 and the composite  $\text{Spin}(10) \times U(1)$  gauge measure, satisfies full reflection positivity along the soldered observer clock  $\tau_\iota$  of Definition 5.11. The residual gap is organized by mirror parity: the 32 EIII directions split into 28 massive coset scalars (covered semiclassically by Theorem 6.15) and 4 massless Goldstones. We conjecture that the massless modes are carried by the  $\sigma$ -invariant sector of  $\hat{Q}$ —mirror partners and their Goldstone doubles are quotient-non-observable (Theorem 5.4)—while the  $\sigma$ -anti-invariant observer sector supplies the observable matter and clock. Full OS reconstruction [9] then requires only the anti-invariant measure factor; the  $\sigma$ -invariant reservoir is a mirror-shadow sector, not an independent postulate.*

### 6.3 Gauge, anomalies, gravity

**Theorem 6.18** (Anomaly cancellation). *For one embedded **27**: cubic  $16 - 80 + 64 = 0$ ; mixed coefficient  $-1$  cancelled by Green–Schwarz [8]. With  $n_F = 3$  replicas of the full **27** arithmetic, anomalies cancel family-wise.*

**Theorem 6.19** ( $b_0 = 12$ ). *With  $C_A(\text{Spin}(10)) = 8$ ,  $T(\mathbf{16}) = 2$ ,  $n_F = 3$ ,  $n_S = 28$ :  $b_0 = (88 - 24 - 28)/3 = 12$ .*

**Theorem 6.20** (Induced Planck mass). *Sakharov induction from 28 massive coset scalars gives  $M_{\text{Pl}}^2 \propto 28\mu^2/(6 \cdot 16\pi^2) \log(\Lambda^2/\mu^2)$ .*

**Definition 6.21** (Dimensionless Newton coupling). At RG scale  $k$ , define

$$\kappa(k) \equiv \frac{k^2}{M_{\text{Pl}}^2(k)}, \quad (6.3)$$

with  $M_{\text{Pl}}(k)$  the running scale induced by the Sakharov flow of Theorem 6.20.

**Theorem 6.22** (Gravitational fixed point). *Assume: (i) Einstein–Hilbert truncation with coupling  $M_{\text{Pl}}^2(k)$ ; (ii) one-loop contribution from the 28 massive coset scalars of Theorem 6.20; (iii) higher-curvature operators and graviton loops neglected; (iv) anomalous dimension  $\eta_{M_{\text{Pl}}} = -(5/48\pi^2)\kappa$  in this truncation. Then*

$$\beta_\kappa \equiv k \frac{d\kappa}{dk} = 2\kappa - \frac{5}{48\pi^2} \kappa^2, \quad (6.4)$$

with  $\kappa_\star = 96\pi^2/5$  and  $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$ .

**Definition 6.23** (Cosmological coupling). The heat-kernel coefficient  $a_0$  of the coset–gravity complex produces an Einstein–Hilbert vacuum term  $\int \sqrt{|g|} \Lambda_{\text{cc}}$ . Treat  $\Lambda_{\text{cc}}$  as an independent dimension-4 coupling in the effective action, not fixed by  $\mu$  or  $M_{\text{Pl}}$ .

**Theorem 6.24** (Vacuum energy RG relevance). *In the same truncation as Theorem 6.20, the  $a_0$  term is scheme-dependent and does not cancel against the massive-mode  $a_2$  sum. Under canonical scaling,*

$$\beta_{\Lambda_{\text{cc}}} = -4\Lambda_{\text{cc}} + \mathcal{O}(\text{loops}), \quad (6.5)$$

so  $\Lambda_{\text{cc}}$  is a relevant direction: a UV fixed point requires  $\Lambda_{\text{cc},\star}$  as input, not as output of void and mirror alone.

**Proposition 6.25** (No-go: CC not from void and mirror alone). *The cosmological constant problem—explaining  $\Lambda_{\text{obs}} \sim (10^{-3} \text{ eV})^4$  without fine-tuning—cannot be resolved from the void and mirror axioms and derived  $E_6$  dynamics alone. Observable configurations lie in  $H^\bullet(\hat{Q}, V_{16})^{\sigma=-1}$  (Definition 6.1); the heat-kernel  $a_0$  vacuum energy and Theorem 6.24 act on the full quotient  $\hat{Q}$ , not only on the anti-invariant sector. Without a sequestering mechanism that routes  $\Lambda_{\text{cc}}$  into the non-observable  $\sigma$ -invariant sector (Theorem 5.4),  $\Lambda_{\text{cc}}$  remains a tuned environmental coupling in the observer sector.*

**Conjecture 6.26** (CC sequestering at  $\kappa_\star$ ). *At the gravitational fixed point  $\kappa_\star = 96\pi^2/5$  (Theorem 6.22), the induced vacuum energy of the coset  $\sigma$ -model flows into the  $\sigma$ -invariant (mirror-partner) sector of  $\hat{Q}$ . By Theorem 5.4,  $\sigma$ -invariant modes are quotient-non-observable; sequestered vacuum energy does not appear in  $\Pi_{\sigma=-1}$ -projected observables, yielding  $\Lambda_{\text{obs}} \approx 0$  without tuning the anti-invariant sector.*

*Remark 6.27* (Observer-local FRG). Mirror collapse  $\Pi_{\sigma=-1}$  (Definition 5.3) splits the effective action into  $\sigma = \pm 1$  sectors. An *observer-local* FRG is the flow restricted to anti-invariant modes along a lifetime chart  $g_\iota(\tau_\iota)$  (Definition 5.11).  $\sigma$ -invariant (mirror-partner) modes are non-observable but may back-react on marginal couplings—notably  $\Lambda_{\text{cc}}$ —without entering the observable Hilbert space.

Collect dimensionless couplings  $(\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\text{cc}})$  with  $\xi = k^2/f^2$ ,  $\hat{g}^2 = k^2 g^2/f^2$ ,  $\hat{y}^2 = k^2 y^2/f^2$ ,  $\kappa$  as in (6.3), and  $\Lambda_{\text{cc}}$  as in Definition 6.23. In the extended one-loop coset–gauge–gravity truncation (still neglecting  $R^2$ , graviton loops, and higher curvature),

$$\begin{aligned} \beta_\xi &= 2\xi, & \beta_{\hat{g}^2} &= 2\hat{g}^2 + \frac{b_0}{48\pi^2}\hat{g}^4, & \beta_{\hat{y}^2} &= 2\hat{y}^2 + \mathcal{O}(\hat{g}^4, \hat{g}^2\hat{y}^2), \\ \beta_\kappa &= 2\kappa - \frac{5}{48\pi^2}\kappa^2, & \beta_{\Lambda_{\text{cc}}} &= -4\Lambda_{\text{cc}} + \mathcal{O}(\text{loops}). \end{aligned} \tag{6.6}$$

**Proposition 6.28** (Jacobian in extended one-loop truncation). *At the fixed point  $(\xi_\star, \hat{g}_\star^2, \hat{y}_\star^2, \kappa_\star, \Lambda_{\text{cc},\star}) = (\xi_\star, 0, 0, 96\pi^2/5, 0)$ , the stability matrix  $J = \partial\beta_i/\partial g_j$  has diagonal spectrum  $\text{eig}(J) = \{+2, +2, +2, -2, -4\}$ . With  $\theta_i = -\text{eig}(\partial\beta_i/\partial g_i)$ , exactly two directions are UV-relevant:  $\kappa$  ( $\theta_\kappa = +2$ ) and  $\Lambda_{\text{cc}}$  ( $\theta_\Lambda = +4$ ). Details in Appendix A.3.*

**Conjecture 6.29** (Graviton/ $R^2$  persistence of  $\kappa_\star$ ). *Beyond the one-loop truncation of Theorem 6.22, include  $R^2$ ,  $R_{\mu\nu}^2$ , and graviton self-interaction in curved-background FRG. The fixed point  $\kappa_\star = 96\pi^2/5$  with  $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$  persists if  $\sigma$ -invariant gravitational modes decouple from the observer-local flow on  $H^1(\hat{Q}, V_{16})^{\sigma=-1}$  (Remark 6.27). Sequestered  $\sigma = +1$  contributions may modify  $\beta_{\Lambda_{\text{cc}}}$  (Conjecture 6.26) but leave  $\beta_\kappa$  unchanged at leading order. Because  $\kappa_\star \approx 189 \gg 1$ , such persistence is not a consequence of void and mirror; it requires nonperturbative FRG input.*

## 6.4 GUT breaking from observer collapse

The heterotic product mirror supplies exact  $\mathbf{16}/\overline{\mathbf{16}}$  duality at  $M^7(\mathcal{V})$ ; observer collapse  $\Pi_{\sigma=-1}$  (Definition 5.3) breaks this pairing and selects chiral descent directions. Symmetry breaking below is gradient flow of  $V$  on  $\iota$ -restricted coset directions, functorially descended from the global EIII minimum above (Definition 5.1).

**Lemma 6.30** (Hessian sector decomposition). *Let  $\Phi_\star \in \mathcal{M}$  be the rank-one EIII vacuum (Theorem 4.5) and write fluctuations  $\delta\Phi = \delta\Phi_{\mathbf{1}} + \delta\Phi_{\mathbf{10}} + \delta\Phi_{\mathbf{16}}$  under the capacity partition  $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$  (Theorem 4.3). On the coset  $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ , the Hessian  $\text{Hess } V(\Phi_\star)$  decomposes  $H$ -equivariantly into  $\mathbf{1}$ ,  $\mathbf{10}$ , and  $\mathbf{16} \oplus \overline{\mathbf{16}}$  blocks. Restricting to the observer diagonal  $\mathfrak{d} = \text{span}\{e_1, e_2, e_3\}$  (Definition 5.5), one obtains a further  $\iota$ -sector split*

$$\text{Hess } V(\Phi_\star)|_{\mathfrak{d}} = \bigoplus_{\iota=1}^3 \text{Hess}_\iota V(\Phi_\star), \tag{6.7}$$

with  $\text{Hess}_\iota$  acting on the  $\iota$ -cell of  $H^1(\hat{Q}, V_{16})^{\sigma=-1, \iota}$ . At tree level  $\text{Hess } V(\Phi_\star) = 0$  on  $\mathfrak{m}$ : the 32 coset directions are exact Goldstones; unstable modes enter only after Coleman–Weinberg radiative correction [10].

**Proposition 6.31** (Chiral breaking directions). *The observer projector  $\Pi_{\sigma=-1}$  selects breaking along  $\sigma$ -anti-invariant coset directions in  $H^1(\hat{Q}, V_{16})^{\sigma=-1}$  and suppresses the mirror-symmetric  $\overline{\mathbf{16}}$  partner. A vacuum-expectation trajectory  $g_\iota(\tau)$  solving (5.8) therefore breaks  $H = \text{Spin}(10) \times U(1)$  along chiral  $\mathbf{16}$ -type fluctuations tied to slot  $\iota$ , not along a simultaneous  $\mathbf{16} \oplus \overline{\mathbf{16}}$  pairing. Duality is exact at the heterotic rung (Theorem 5.4); physical breaking is the localized violation induced by collapse.*

*Proof sketch.*  $\Pi_{\sigma=-1} = \frac{1}{2}(\text{id} - \sigma)$  annihilates  $\sigma$ -invariant modes. On  $\mathfrak{m}$  the mirror involution exchanges  $\mathbf{16}_{-3}$  and  $\overline{\mathbf{16}}_{+3}$  (Proposition 2.7); anti-invariant fluctuations lie in  $\mathbf{16}_{-3}$  modulo the  $\iota$ -diagonal restriction. The lifetime flow (5.8) restricts  $\nabla V$  to the  $\iota$ -sector, breaking product mirror symmetry while preserving capacity balance on the global datum  $\mathfrak{U}^*$ .  $\square$

**Theorem 6.32** (First breaking, conditional). *Assume the EIII vacuum  $\Phi_*$  of Theorem 4.5, observer collapse on  $\hat{Q}$ , and a fixed observer index  $\iota \in \mathcal{I}$ . If radiative corrections lift the  $\mathbf{16}$ -sector Hessian with a single unstable mode per  $\iota$ -cell, then gradient flow of  $V$  on the  $\iota$ -restricted coset triggers the first step*

$$E_6 \longrightarrow \text{Spin}(10) \times U(1) \quad (6.8)$$

*along the rank-one orbit of  $\mathcal{M}$ , with  $U(1)$  charge aligned to the  $\iota$ -diagonal of  $\mathfrak{d}$ . The residual stabilizer  $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$  is the symmetry group of Theorem 4.7(iii).*

*Proof sketch.* Rank-one EIII fixes  $\mathcal{M} = E_6/(\text{Spin}(10) \times U(1))$  globally (Theorem 4.5). An  $\iota$ -localized unstable fluctuation in  $\mathbf{16}_{-3}$  is an adjoint-direction Goldstone of  $E_6/H$ ; flow along that mode restores a  $\text{Spin}(10)$ -invariant submanifold while breaking the remaining  $U(1)_\iota \subset U(1)$  tied to the chosen diagonal. The branching matches the standard  $\mathbf{78} = \mathbf{45} \oplus \mathbf{1} \oplus \mathbf{16} \oplus \overline{\mathbf{16}}$  pattern at the  $\text{Spin}(10)$ -invariant vacuum [12, 13].  $\square$

**Conjecture 6.33** (Derived GUT–Higgs chain). *Sequential  $\iota$ -restricted gradient flow on the 28 radiatively massive coset directions (Coleman–Weinberg, as in Theorem 6.15) triggers the full chain*

$$E_6 \rightarrow \text{Spin}(10) \times U(1) \rightarrow SU(5) \rightarrow SU(3)_c \times SU(2)_L \times U(1)_Y \rightarrow SU(3)_c \times U(1)_{\text{em}} \quad (6.9)$$

*with observable Higgs content  $\mathbf{45}_H$ ,  $\mathbf{126}_H$ ,  $\mathbf{10}_H$  realized as unstable Hess  $V$  directions in the  $\mathbf{27}$ -sector of Theorem 4.3, filtered through  $\Pi_{\sigma=-1}$ . Yukawa matrices  $y_{ij}$  arise from cubic overlap integrals*

$$y_{ij} \propto \int_{\hat{Q}} \omega_i \wedge \omega_j \wedge \Phi_{\text{Higgs}}, \quad (6.10)$$

*of observer modes  $|\omega_i\rangle$  on  $\hat{Q}$  (Definition 5.6) against Higgs fluctuations  $\Phi_{\text{Higgs}}$  drawn from the unstable Hessian spectrum.*

**Conjecture 6.34** (GUT breaking). *Gradient flow of  $V$  on  $\iota$ -restricted coset directions triggers (6.9) with Higgs content  $\mathbf{45}_H$ ,  $\mathbf{126}_H$ ,  $\mathbf{10}_H$ ; Conjecture 6.33 supplies the observer-derived mechanism (Lemma 6.30, Proposition 6.31, Theorem 6.32).*

*Remark 6.35* (As above, so below). Global capacity balance fixes the EIII minimum  $\Phi_* \in \mathcal{M}$  above (Theorem 4.5); observer collapse localizes breaking below to the  $\iota$ -sector lifetime chart (Definition 5.11). The correspondence (5.1) identifies the profinite universe datum  $\mathfrak{U}^*$  with fixed-point collapse on  $\hat{Q}$ : the same potential  $V$  governs the global vacuum and the slot-resolved descent trajectories  $g_\iota$ . What is proved globally on  $\mathfrak{U}^*$ —the 1|10|16 split, rank-one EIII, three observer slots—appears locally as chiral gauge breaking in slot  $\iota$ , without importing mirror-symmetric partners.

## 6.5 Flavon hierarchy from observer capacity flow

Assume Conjecture 6.34. The flavour group  $SU(3)_F \subset \text{Spin}(10)$  acts on the three chiral  $\mathbf{16}$ s; each observer index  $\iota \in \mathcal{I}$  labels one diagonal direction of  $\mathfrak{d} \subset J_3(\mathbb{O})$  and hence one  $SU(3)_F$  flavour axis.

**Proposition 6.36** (Flavon sector from observer indices). *The three observer slots of Theorem 6.4 index three  $SU(3)_F$  breaking stages. Along the lifetime chart  $g_i(\tau_i)$  (Definition 5.11), the capacity flow  $\omega_i^*(\tau) = \mathcal{C}_i(\tau)$  of Definition 5.12 records which flavon direction is active at time  $\tau$  within slot  $i$ . Mirror stabilisation fixes  $n_F = 3$  but not, by itself, Yukawa eigenvalues or mixing matrices.*

**Conjecture 6.37** (Flavon hierarchy, P3'). *Sequential  $SU(3)_F \rightarrow SU(2)_F \rightarrow U(1)_F$  breaking is driven by the observer-indexed capacity RG flow  $\omega_i^*(\tau)$  on the flavon potential  $V_F(\phi_1, \phi_2, \phi_3)$ , with  $\phi_i$  the flavon along diagonal  $e_i$ . The ordering  $\langle \phi_3 \rangle \gg \langle \phi_2 \rangle \gg \langle \phi_1 \rangle$  follows from the capacity weights  $(1/27, 10/27, 16/27)$  of Theorem 4.3 descended through the above–below correspondence (Proposition 5.2), yielding Yukawa hierarchy and CKM/PMNS mixing without an independent flavon postulate.*

## 7 Predictions and Experimental Contact

SJET is not complete until its outputs confront measurement. This section records **what is already fixed** by void, mirror, and observer collapse, **what follows conditionally** from capacity descent and flavon flow, and **which near-term experiments** (2025–2030) can falsify or corroborate each tier. All comparisons use public results as of mid-2026; cited values are experimental central values unless noted.

### 7.1 Prediction protocol

**Definition 7.1** (Prediction tiers). **T1. Exact** — integer or rational consequence of proved theorems (no free parameters).

**T2. Leading** — order-of-magnitude from capacity weights  $(1/27, 10/27, 16/27)$  and observer index  $i$ .

**T3. Conditional** — follows if Conjectures 6.33, 6.37, 6.26 hold; falsifiable but not yet proved from axioms alone.

**Definition 7.2** (Contact map). A *contact map* assigns each prediction  $P$  to an observable  $\mathcal{O}$  and experiment  $E$  with tolerance  $\delta$ :

$$\mathcal{C}(P) = (\mathcal{O}, E, \delta), \quad \text{falsified if } |\mathcal{O}_{\text{exp}} - \mathcal{O}_{\text{SJET}}| > \delta. \quad (7.1)$$

### 7.2 Tier T1: exact predictions

| Quantity                       | SJET value  | Source    | Experiment (2026)         |
|--------------------------------|-------------|-----------|---------------------------|
| Family count $n_F$             | 3           | Thm. 6.4  | No 4th gen (LHC)          |
| $\beta$ -function $b_0$        | 12          | Thm. 6.19 | SM 1-loop $\beta_\lambda$ |
| Coset scalars $n_S$            | 28          | Thm. 6.20 | EWSB scalar count         |
| Mixed GS coeff. (1 fam.)       | −1          | Thm. 6.18 | String/GS sign            |
| $\kappa_\star$                 | $96\pi^2/5$ | Thm. 6.22 | FRG benchmark             |
| Observer slots $ \mathcal{I} $ | 3           | Thm. 5.9  | —                         |

Table 1: Exact (T1) predictions.  $\kappa_\star \approx 189.6$  is dimensionless in the stated truncation; physical scale identification is not T1.

**Proposition 7.3** (No fourth generation). *The Lefschetz index  $\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = -3$  fixes  $|\mathcal{I}| = 3$  exactly. A fourth chiral family requires a fourth  $\sigma$ -anti-invariant 1-class on  $\hat{Q}$ , contradicting Theorem 5.9. SJET predicts **no** sequential fourth generation at any energy reachable without changing the heterotic quotient structure.*

*Remark 7.4* (Muon  $g-2$  contact, T1). Fermilab Muon  $g-2$  (June 2025 final run) reports  $a_\mu = 0.001\,165\,920\,705(205)$  [14]. SJET does not fix  $a_\mu$  at T1: the 28 coset scalars and three families contribute through *unspecified* masses. The T1 prediction is **structural**: new physics from a fourth family or extra mirror-observable chiral families is excluded; any residual anomaly must arise from the **27** sector with  $n_F = 3$  and  $b_0 = 12$ .

### 7.3 Tier T2: leading-order spectrum

Capacity weights  $(m_1, m_{10}, m_{16}) = (1/27, 10/27, 16/27)$  descend to observer indices  $\iota = 1, 2, 3$  on  $\mathfrak{d}$ .

**Proposition 7.5** (Generation ordering). *Under the flavon sector map of Proposition 6.36, observer index  $\iota$  carries capacity weight  $\omega_\iota^* \propto w_\iota$  with*

$$w_1 : w_2 : w_3 = 1 : 10 : 16. \quad (7.2)$$

*Leading-order Yukawa hierarchies satisfy  $y_\iota \propto \sqrt{w_\iota}$  and  $m_\iota \propto w_\iota$  at fixed Higgs VEV.*

*Prediction 7.6* (Normal neutrino mass ordering). **T2**. The heaviest neutrino mass eigenstate is  $m_3$  (normal ordering):  $w_3 > w_2 > w_1$  implies  $m_3 > m_2 > m_1$  before RG corrections. **Contact**: JUNO mass-ordering determination (2026–2028);  $\mathcal{C}(\text{NO}) = (\text{sign}(\Delta m_{31}^2 - \Delta m_{21}^2), \text{JUNO}, \text{sign test})$ .

*Prediction 7.7* (Mass-squared ratio target). **T2 (leading)**. Capacity descent suggests

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} \sim \frac{w_3}{w_1} = 16 \quad (\text{order of magnitude, pre-RG}). \quad (7.3)$$

PDG 2025 gives  $\Delta m_{31}^2 / \Delta m_{21}^2 \approx 33$  [15]. SJET targets the same decade at leading order; refinement requires Conjecture 6.37 (two-loop capacity RG on  $V_F$ ). **Falsifier**: inverted ordering; ratio  $\ll 3$  or  $\gg 10^3$  after sub-percent JUNO oscillation fits.

*Prediction 7.8* (Sector mass ratios). **T2**. Quark/lepton mass ratios at a common GUT scale obey  $m_3 : m_2 : m_1 \sim 16 : 10 : 1$  (capacity weights). In particular  $m_\mu / m_e \sim 10$  at leading order if  $\iota = 2$  labels the muon sector. **Contact**: PDG mass ratios;  $\mu$   $g-2$  constrains  $\iota = 2$  loop contributions from the 10/27 sector.

### 7.4 Tier T3: conditional predictions

*Prediction 7.9* (Observable cosmological constant). **T3**. If Conjecture 6.26 holds at  $\kappa_\star$ ,

$$\Lambda_{\text{obs}} \approx 0 \quad (7.4)$$

in the  $\sigma = -1$  observer sector, with environmental vacuum energy stored in  $H^\bullet(\hat{Q})^{\sigma=+1}$ . **Contact**: DESI/JWST dark-energy equation of state  $w \rightarrow -1$  with no fine-tuned drift;  $\mathcal{C} = (w, \text{DESI}, |w+1| < 0.05)$ . Does *not* by itself resolve the Hubble tension; a non-zero  $\Lambda_{\text{obs}}$  in the sequestered framework would falsify T3.

*Prediction 7.10* (GUT-scale unification). **T3.** Under Conjecture 6.33 with Spin(10) survival to  $M_{\text{GUT}}$ ,

$$\sin^2 \theta_W(M_{\text{GUT}}) = \frac{3}{8}, \quad \alpha_{\text{em}}^{-1}(M_{\text{GUT}}) \approx 24, \quad (7.5)$$

with  $b_0 = 12$  fixing the Spin(10) normalization of Thm. 6.19. **Contact:** high-scale coupling unification fits (SUSY or non-SUSY); deviation of  $b_0$  from 12 shifts unification scale.

*Prediction 7.11* (Mirror-partner null). **T3.** No observable process exhibits exact  $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$  symmetry at the amplitude level: mirror partners are  $\sigma$ -invariant and quotient-non-observable (Theorem 5.4). **Contact:** searches for wrong-sign heavy-lepton or mirror-coupled  $\mathbf{16}$  partners at colliders; any sustained mirror-symmetric rate at production level would falsify the observer-collapse sector definition.

## 7.5 Near-term experimental map (2025–2030)

| Experiment         | Observable          | Tier  | SJET expectation                                         |
|--------------------|---------------------|-------|----------------------------------------------------------|
| JUNO               | Mass ordering       | T2    | Normal ( $m_3$ heaviest)                                 |
| JUNO               | $\Delta m^2$ ratios | T2    | $\Delta m_{31}^2 / \Delta m_{21}^2 \sim \mathcal{O}(10)$ |
| Muon $g-2$         | $a_\mu$             | T1/T2 | No 4th family; 3-family + 28 scalars                     |
| LHC/FCC            | 4th generation      | T1    | Excluded structurally                                    |
| DESI Phase II      | $w(z)$              | T3    | $w \approx -1$ if CC sequestered                         |
| KATRIN / Project 8 | $m_\beta$           | T2    | Consistent with NO + capacity                            |
| HL-LHC             | Spin(10) Higgs      | T3    | <b>45, 126, 10</b> pattern                               |

Table 2: Near-term experimental contact map.

## 7.6 Development program toward precision match

To move predictions from leading order to *sub-percent* agreement with JUNO,  $g-2$ , and cosmology, the theory must close three derivation gaps in order:

**D1. Yukawa from overlaps** — evaluate (6.10) on  $\hat{Q}$  with explicit  $\Phi_{\text{Higgs}}$  from Hess  $V$  (Conjecture 6.33); produces  $y_{ij}$  and  $m_\ell$  without postulates.

**D2. Flavon RG** — prove Conjecture 6.37 to refine (7.3) from 16 to the PDG value  $\approx 33$ .

**D3. Scale identification** — map  $\kappa_\star$  and  $\mu$  from Theorem 6.20 to  $M_{\text{Pl}}$  and electroweak scale using observer-local FRG (Remark 6.27).

*Remark 7.12* (Falsifiability standard). A prediction is *on programme* when it has a contact map (7.1) and a named derivation step D1–D3. Until D1–D3 close, T2 predictions are order-of-magnitude targets, not precision fits; T1 predictions are the primary falsifiers for new physics searches.

## 8 Conclusions

SJET builds the universe from the mirror:

$$\mathcal{V} \xrightarrow{\text{M}} \dots \xrightarrow{\text{M}} \mathfrak{U}^\star, \quad \mathfrak{U}^\star = \text{C}^\infty(\mathcal{V}). \quad (8.1)$$

Observers arise at the heterotic rung as collapsed fixed-point states:

$$\iota \mapsto |\omega_\iota\rangle \in H^1(\hat{Q}, V_{16})^{\sigma=-1}, \quad g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota. \quad (8.2)$$

Only two axioms are primitive. Duality, observer multiplicity, and the index–lifetime map are **derived** from  $E_8 \times E_8$  collapse and the Lefschetz fixed-point theorem.

## 8.1 Epistemic summary

| Status       | Content                                                                                                                                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axioms       | void, mirror                                                                                                                                                                                                                                                                                    |
| Theorems     | rung table, $E_8$ , $E_8 \times E_8$ , $E_6$ , partition, vacuum, duality, observer fixed points, chirality derived, index–lifetime map, soldering derived, soldering clock, anomalies, $b_0 = 12$ , $\kappa_\star$ (one-loop trunc.), CC RG relevance, semiclassical RP, extended FRG Jacobian |
| Definitions  | above–below, observer, lifetime chart, $H_2(\mathbb{H})$ block, physical sector, $\kappa$ , $\Lambda_{\text{cc}}$                                                                                                                                                                               |
| Propositions | CC no-go (void+mirror alone), flavon sector from $\iota$                                                                                                                                                                                                                                        |
| Conjectures  | lifetime uniqueness, soldering uniqueness, full RP, GUT breaking, CC sequestering (Conj. 6.26), $\kappa_\star$ graviton/ $R^2$ persistence (Conj. 6.29), flavon hierarchy P3' (Conj. 6.37)                                                                                                      |

## 8.2 Prediction development

Section 7 maps SJET to 2025–2030 experiments. **Immediate falsifiers (T1):**  $n_F = 3$ ,  $b_0 = 12$ , no fourth generation. **Near-term targets (T2):** normal neutrino ordering and  $\Delta m_{31}^2 / \Delta m_{21}^2 \sim \mathcal{O}(10)$  from capacity weights 1 : 10 : 16. **Conditional (T3):**  $\Lambda_{\text{obs}} \approx 0$ ,  $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$ . Closing derivation steps **D1–D3** (Section 7.6) is required for sub-percent agreement with JUNO and Muon  $g-2$ .

## 8.3 Open problems

- O1.** Prove Conjecture 5.16 (functorial lifetime chart).
- O2.** Prove Conjecture 6.13 (functorial soldering; Theorem 6.10 derives  $d = 4$  conditionally).
- O3.** Prove Conjecture 6.34 from  $V$ .
- O4.** Prove Conjecture 6.17.
- O5. Partially addressed (O4):** Conjecture 6.29 frames graviton/ $R^2$  persistence of  $\kappa_\star$  via observer-local FRG with  $\sigma$ -invariant decoupling; *still open*: curved-background FRG proof.
- O6. Partially addressed (O5):** Proposition 6.25 and Conjecture 6.26 route  $\Lambda_{\text{cc}}$  to the  $\sigma = +1$  sector at  $\kappa_\star$  (Theorem 5.4); *still open*: derive sequestering from the effective action.
- O7. Partially addressed (P3'):** Proposition 6.36 and Conjecture 6.37 reduce flavon hierarchy to capacity flow  $\omega_\iota^\star(\tau)$  on  $SU(3)_F$ ; *still open*: prove ordering and mixing matrices from  $V_F$ .

## 8.4 Archive

The prior modular manuscript (39 pages, verification ledger, axiomatic elimination program) is preserved in `old/sjet-stack/`.

## 9 Epistemic Gap Ledger

The prior modular manuscript recorded interim *physical postulates* P1–P5 and a tuned cosmological constant CC. After the observer theory of Section 5—in particular the as above, so below correspondence (Definition 5.1)—several gaps close by identifying global capacity data with localized collapse on  $\hat{Q}$  at  $M^7(\mathcal{V})$ . Table 3 records status *after* observer theory; routes name the above–below mechanism.

*Remark 9.1* (Ledger reading). *Eliminated* means the former postulate is replaced by a theorem or definition from void, mirror, and capacity together with observer collapse. *Derived* means the structural content is supplied by the climb with a named residual open step. *Conjecture* and *Open* mark honest remaining inputs to a conditional contact with four-dimensional quantum field theory.

## A Albert algebra and mirror quotient

### A.1 Albert model

Elements of  $J_3(\mathbb{O})$  are  $3 \times 3$  Hermitian octonionic matrices

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbb{R}, \quad x_i \in \mathbb{O}. \quad (\text{A.1})$$

The sharp product  $X^\#$  and cubic norm  $N(X) = \det X$  define the  $E_6$  structure group [5, 6].

### A.2 Mirror quotient cohomology

On the stabilized tower  $\mathcal{T}_7$  at  $M^7(\mathcal{V})$ , let  $\sigma$  act on the rank-3 diagonal  $\mathfrak{d}$  by  $\sigma(e_i) = -e_i$  with triality rotation on the three **16**-cells  $(C_1, C_2, C_3)$ .

**Lemma A.1** (Cellular observer count). *With  $\mathcal{F}_{16}$  the **16**-cellular cosheaf on  $\mathcal{T}_7$ ,*

$$\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=+1} = 0, \quad \dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3. \quad (\text{A.2})$$

*Each  $\sigma = -1$  class is one observer index  $\iota \in \{1, 2, 3\}$ .*

*Proof.* Cellular 1-cochains are triples  $(\phi_1, \phi_2, \phi_3) \in (\mathbb{C}^{16})^3$  with cocycle constraint  $\phi_1 + \phi_2 + \phi_3 = 0$  at the **10**-hub. Coboundaries are diagonals  $(\psi, \psi, \psi)$ . The quotient is two copies of  $\mathbb{C}^{16}$  as a  $\sigma$ -module; triality with sign leaves three anti-invariant classes, one per diagonal direction  $e_i$ .  $\square$

**Proposition A.2** (Stone lift). *The profinite lift to  $Q$  identifies  $H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} \cong H^1(Q, V_{16})^{\sigma=-1}$ .*

**Theorem A.3** (Mirror-quotient Lefschetz index). *With  $\bar{\partial}_{V_{16}}$  on  $(Q, V_{16})$  and antiholomorphic  $\sigma$ ,*

$$\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = \text{Tr}_{H^1(Q, V_{16})}(\sigma) = -3, \quad \dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3. \quad (\text{A.3})$$

*This is Theorem 5.9; each unit contribution is one observer fixed-point slot.*

*Proof.* Lemma A.1, Proposition A.2, and the Atiyah–Bott holomorphic Lefschetz formula [7]. For a  $\mathbb{Z}_2$  quotient by an antiholomorphic involution,  $H^1(\hat{Q}, V_{16}) = H^1(Q, V_{16})^{\sigma=-1}$ .  $\square$

### A.3 Extended FRG Jacobian

At the fixed point of (6.6) with  $\hat{g}_\star^2 = \hat{y}_\star^2 = 0$ , loop corrections to each  $\beta_i$  are proportional to a vanishing coupling, so  $J = \partial\beta_i/\partial g_j$  is diagonal:

$$\partial_\xi\beta_\xi = +2, \quad \partial_{\hat{g}^2}\beta_{\hat{g}^2} = +2, \quad \partial_{\hat{y}^2}\beta_{\hat{y}^2} = +2, \quad \partial_\kappa\beta_\kappa = -2, \quad \partial_{\Lambda_{\text{cc}}}\beta_{\Lambda_{\text{cc}}} = -4. \quad (\text{A.4})$$

Hence  $\text{eig}(J) = \{+2, +2, +2, -2, -4\}$  and, with  $\theta = -\text{eig}(J)$ , the UV-relevant directions are  $\kappa$  and  $\Lambda_{\text{cc}}$  only. The coset scalar block  $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$  contributes no additional relevant directions on the EIII vacuum. This is Proposition 6.28; Conjecture 6.26 targets the  $\Lambda_{\text{cc}}$  row by  $\sigma$ -sector decoupling, and Conjecture 6.29 the  $\kappa$  row under graviton/ $R^2$  extension.

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| ID  | Former gap                            | Status            | Route (observer / above–below)                                                                                                                                                                                                                                                                             |
|-----|---------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1  | Lorentzian soldering, $d = 4$         | <b>Derived</b>    | <i>Below:</i> mirror-fixed $\mathbb{H} \subset \mathbb{O}$ at Hurwitz terminal ( $n = 4$ ) supplies $H_2(\mathbb{H})$ and the clock block (Definition 6.7); <i>above:</i> quaternionic sector of $\mathfrak{U}^*$ capacity flow. Functorial uniqueness remains open (Open Problem <b>O2</b> in Section 8). |
| P2  | Chiral matter                         | <b>Eliminated</b> | <i>Below:</i> physical sector $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ and observer collapse $\Pi_{\sigma=-1}$ (Definitions 6.1, 5.6); chirality is $\sigma$ -anti-invariance, not postulated content. <i>Above:</i> <b>16</b> bundle on the 1 10 16 partition.                                                  |
| P3  | Three families                        | <b>Eliminated</b> | <i>Below:</i> Lefschetz fixed-point theorem, $\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3$ (Theorem 5.9); minimal heterotic rung (Proposition 3.7). <i>Above:</i> three diagonal directions of $\mathfrak{d} \subset J_3(\mathbb{O})$ and capacity weights $(1/27, 10/27, 16/27)$ (Proposition 5.2).         |
| P3' | Flavon hierarchy, CKM/PMNS            | <b>Conjecture</b> | <i>Below:</i> capacity flow $\omega_t^*(\tau)$ on $SU(3)_F$ indexed by observer slots (Proposition 6.36, Conjecture 6.37); <i>above:</i> $(1/27, 10/27, 16/27)$ weights. VEV ordering and mixing matrices remain open.                                                                                     |
| P4  | GUT–Higgs and Yukawa                  | <b>Conjecture</b> | <i>Above:</i> gradient flow of $V$ on EIII (Conjecture 6.33); <i>below:</i> $\iota$ -restricted chiral breaking via $\Pi_{\sigma=-1}$ (Theorem 6.32, Section 6.4).                                                                                                                                         |
| P5  | Nonperturbative reflection positivity | <b>Conjecture</b> | Semiclassical RP on EIII is <b>derived</b> (Theorem 6.15); full OS reconstruction remains Conjecture 6.17. The below-level gap is unitary Lorentzian completion beyond the soldered semiclassical regime.                                                                                                  |
| CC  | Cosmological constant                 | <b>Conjecture</b> | Proposition 6.25: not derivable from void–mirror alone. Target: Conjecture 6.26 routes $\Lambda_{cc}$ to the $\sigma = +1$ mirror-partner sector at $\kappa_*$ (Theorem 5.4).                                                                                                                              |

Table 3: Epistemic elimination ledger after observer theory. Only void and mirror are primitive; P2 and P3 retire via heterotic collapse and fixed points.